

Status of the Gunter & Scott 1990 Development in Lean

Measured on main at 5a1d8bc. Build 1372 jobs, 0 errors, 0 warnings. 124 modules, 46,097 lines, 2,119 theorems.

1. Extractions — what the paper contains

#	Kind	Count
1	Definitions	≈13
2	Numbered theorems	16
3	Numbered lemmas	13
4	Numbered propositions	1
5	— numbered results, total	30
6	— the conjuncts those 30 decompose into	93
7	Prose statements	146
8	Examples	not measured

Numbers 1–30 are contiguous, no gap and no repeat, verified by extracting every (Theorem\|Lemma\|Proposition\|Corollary) N heading from the full text.

Two rows are soft. Row 1 has stood at “≈13” since r0031 and has never been enumerated to an exact figure. Row 8 is a real hole: no round has counted the paper’s examples at all. Figure 4’s element counts (1, 2, 5, 20) were used to adjudicate a §7.4 reading, so at least some examples have been read — but never enumerated. Both should be counted the same way the 30 were: extract the headings from the full text.

2. Represented in Lean

#	Kind	Represented	Of	Not stated
1	Definitions	≈13	≈13	0
2	Theorems	16	16	0
3	Lemmas	13	13	0
4	Propositions	1	1	0
5	Prose statements	86	146	60
6	Examples	—	—	not measured

All 30 numbered results have been stated in Lean since r0048. The 60 unstated prose statements are concentrated rather than spread: 13 of the 26 unstated *properties* are §7 alone.

3. Corrected — a statement that could not be transcribed as printed

Cause on the left, **what we did about it on the right**. Every one has a corrected statement in Lean; the last two columns say whether that corrected statement is proven.

#	Kind	Printing	Reading	Semantic	Corrected	— proven	— unproven
1	Definitions	1	0	1	2	2	0
2	Theorems	1	1	2	4	2	2
3	Lemmas	1	2	0	3	2	1
4	Propositions	0	0	0	0	0	0
5	— total	3	3	3	9	6	3
6	Prose statements	—	—	—	—	—	not classified
7	Examples	—	—	—	—	—	not measured

Nothing is left uncorrected. All 9 have a corrected statement in Lean, and 6 of the 9 are proven in that form. The 3 unproven ones are exactly the 3 unproven results in table 4 — no correction is waiting on a decision, only on a proof.

Column meanings, and every row behind the counts:

Printing error — the printed page is unreadable or garbled; the mathematics is right.

#	Where	What	Corrected to	Proven
1	Lemma 9	the PDF drops every $\otimes$ and every $\perp$, so which operators the laws range over is unreadable	six laws recovered, <code>lem9_1...lem9_6</code> ; items 3 and 5 also carry kernel-checked negations of the printed forms	yes
2	Theorem 14	the list of characterizations is garbled	<code>thm14</code> , both directions	yes
3	§7.4’s worked example	reverses the paper’s own definition — printed $b \vdash a$ where both papers give $a \vdash b$	read in the paper’s own direction; the paper’s element counts 1, 2, 5, 20 select this reading over the rival	yes

Reading error — *ours*. The paper is right; we transcribed something it does not say.

#	Where	What	Corrected to	Proven
1	Theorem 29, sentence 2	we dropped the word “domain” from “any bifinite domain”, which makes the sentence false	<code>Thm29SecondAtDomains</code>	no
2	Lemma 28	we quantified over all U ; the paper states it over its own U	<code>PRep.Lemma28AtU</code> , all nine conjuncts, no hypotheses	yes
3	Lemma 30	same shape — we wrote a universal closure the paper does not claim	<code>Lemma30AtV</code> , now at arity 1	no

Semantic error — the mathematics is wrong as stated.

#	Where	Whose	What	Corrected to	Proven
1	Theorem 26	the paper’s	false for any signature with two or more 0-ary slots, including the paper’s own worked $(2, 0, 0, 0, 0, 0)$	<code>thm26</code> with $hs : \forall i, 0 < s\ i$ — the added binder is the repair	yes
2	Theorem 7	ours	our guard tested compactness where the sentence is about boundedness; on the paper’s own index the join always exists, so it tests neither	<code>StepFunctionsDecidable</code> restated over <code>IsStepEnumeration</code> , existentially quantified, naming no guard	no
3	§7.4’s order relation	the paper’s	the printed relation is not reflexive — $b = (\perp, \emptyset)$ fails $b \vdash b$. It is the strict part, and the order is its reflexive closure, which is the identification §7.4 then performs by hand	reflexive closure taken; <code>Colimit.V</code> built on it, with <code>isoPlus : V $\approx_0$ Plus V</code>	yes

Lemma 9’s two misprinted items (3 and 5) are also individually false as printed; they are counted once under Lemma 9’s printing error, since a single unreadable page produced both.

4. Proof state

#	Kind	Proven	Unproven	Of
1	Definitions	n/a	n/a	—
2	Theorems	14	2	16
3	Lemmas	12	1	13
4	Propositions	1	0	1

#	Kind	Proven	Unproven	Of
5	— numbered results	27	3	30
6	Prose statements	70	16	86 stated
7	Examples	—	—	not measured

A definition is not proven or unproven; the thing that can go wrong is that it is **empty** — a definition nothing satisfies makes every theorem over it vacuous. The measurement is instantiation: **25 structures and classes, 0 never instantiated**. Two mechanisms of vacuity have been found and are now checked for: fields inhabited free by `Classical.dec`, and a conclusion parameter with no hypothesis parameter in its component. The second sweep returned 1 hit in 2,111 declarations, the already-known one.

The three unproven numbered results, and what each still needs:

#	Result	State
1	Theorem 7	sentences 1 and 2 proven; the recursive forms are not
2	Theorem 29	sentence 1 proven; sentence 2 is not
3	Lemma 30	stated at V; arity 1 — one named input left

All three reduce to two statements, and nothing else is outstanding:

#	The unproven statement	Blocks
1	<code>RecursiveNormal</code> for $K(D \rightarrow E)$ — decide whether two basis elements are bounded, and compute where their join is	Theorem 7
2	<code>Thm29Normal</code> at infinite $K(E)$ — realize the type over the tower's image, not the η -image. Finite bases are proven	Theorems 29, Lemma 30

Neither is unsolved mathematics: Gunter & Scott proved both in 1990.

5. Proof work state

#	Kind	Started, left with a sorry
1	Definitions	0
2	Theorems	0
3	Lemmas	0
4	Propositions	0
5	Prose statements	0
6	Examples	0

sorry is 0 across the package, and has been since r0042. So is `axiom`, and so is any constant naming `sorryAx`.

That number does not mean everything is proven, and the difference matters when reading table 4. A `sorry` is a hole in a proof somebody *started*. The three unproven results were not started and left open — each is carried as an explicit hypothesis, so a theorem that needs one takes it as an argument. Nothing is asserted that is not proven, and no `sorryAx` propagates; but no `sorry` count, here or anywhere, can see this work.

Reproducing

```
scripts/counts.sh           # modules, lines, theorems, sorry
scripts/numbered-status.sh  # declarations per numbered result
scripts/a6-env-scan.sh <out> # then a6-summarize.py
scripts/a8-claim-check.sh   # prose staleness against the environment
scripts/compile.sh -r <round> # build
```